Harsh Shroff

Open to Relocation

667-464-5255 harshrofff@gmail.com

linkedin.com/in/harshroff/ github.com/HarshShroff/ harshshroff.github.io/

PROFESSIONAL SUMMARY

Robotics & **AI/ML Engineer** specializing in multi-modal systems integration with hands-on **Boston Dynamics Spot** and **Microsoft HoloLens2** experience. AWS Machine Learning Associate Certified with proven track record leading teams of 5+ engineers to deliver production AI systems achieving **60% efficiency gains** and **92% accuracy** in real-world deployments. Expert in bridging robotics research and commercial applications through advanced computer vision, reinforcement learning, and geospatial AI. Published researcher ready to drive next-generation autonomous systems for leading technology companies.

CORE COMPETENCIES

Robotics & Autonomous Systems: Boston Dynamics Spot Integration, ROS, Multi-modal Navigation, LiDAR, HoloLens 2, Sim2Real Deployment, Human-Robot Teaming, Reinforcement Learning, Computer Vision

AI/ML & Deep Learning: Generative AI (LLMs, RAG, LangChain), PyTorch, TensorFlow, YOLOv8, OpenCV, NLP, Time Series, MLOps (MLflow, Kubeflow), Prompt Engineering

Cloud & Production Systems: AWS (Lambda, EC2, S3, Textract, Bedrock), Databases (PostgreSQL, MongoDB), ETL Pipelines, Docker, Kubernetes, Terraform, CI/CD

Programming & Development: Python, SQL, R, Embedded C, API Development (FastAPI, Flask), Git, Agile

Data Engineering & Visualization: Geospatial Analytics, Tableau, Streamlit, Dash, PostGIS, Spark, Hadoop

PROFESSIONAL EXPERIENCE

AI Researcher - Robotics Systems Integration — **Jun 2025 – Present**

Center for Real-Time Distributed Sensing and Autonomy (CARDS), UMBC

- Engineering cutting-edge human-robot teaming system integrating **Boston Dynamics Spot** with real-time spatial mapping and **Microsoft HoloLens** AR, enabling precise movement mimicking and autonomous collaboration. Video
- Developing advanced multi-modal perception algorithms combining LiDAR, radar, visual, spatial, and gestural inputs for next-generation autonomous systems applications.

ML Engineer - Geospatial AI Systems — **Jun 2024 – May 2025**

VITG Corp., Halethorpe, MD

- Architected and deployed production **Llama LLM-based automation system** on AWS (Lambda, EC2, S3) reducing candidate screening process time by **60%** and streamlining HR workflows for 200+ employee organization.
- Built commercial **geospatial chatbot** using Claude 3.5 and AWS Textract, automating city zoning analysis for urban planners in Hyderabad, processing complex regulatory queries in real-time.
- Automated extraction and analysis of **2,500+ environmental documents** using LLM-OCR pipelines, achieving **40% efficiency improvement** in regulatory compliance and environmental planning workflows.

AI Researcher - Autonomous Robotics — **Mar 2023 – Dec 2024**

Center for Real-Time Distributed Sensing and Autonomy (CARDS), UMBC

- **Led cross-functional team of 5 engineers** developing multi-modal navigation systems for **Boston Dynamics Spot** and Clearpath robots under **US Army Research Lab funding**, advancing sim-to-real deployment capabilities.
- Pioneered gesture-based robot control using **Microsoft HoloLens 2** and Python, enabling intuitive hands-free human-robot collaboration in complex environments—first-of-its-kind integration in academic research.
- Optimized LIDAR-based navigation using **reinforcement learning algorithms**, achieving **20% reduction in navigation errors** and improving real-time pathfinding performance in both simulation and field deployments.

Data Science Intern - Computer Vision — **Aug 2023 – Dec 2023**

Freshwater Institute – The Conservation Fund, Shepherdstown, WV

- Developed production-ready **computer vision system** achieving **92% accuracy** in automated fish fillet color grading using YOLOv8 and OpenCV, deployed on Raspberry Pi for real-time aquaculture quality control.
- Designed and prototyped portable, 3D-printed assessment hardware enabling on-site deployment of AI quality control systems in commercial aquaculture operations.
- Co-authored peer-reviewed research paper published in Journal of Agriculture and Food Research, demonstrating commercial impact of deep learning in precision agriculture.

EDUCATION

Master's in Data Science **Aug 2022 – May 2024**
University of Maryland Baltimore County (UMBC) | ***GPA: 3.8/4.0***
- Relevant Coursework: Natural Language Processing, Machine Learning, Artificial Intelligence, Computer Vision, Big Data Systems, Predictive Analytics, Business Intelligence

Bachelor's in Electronics & Communication Engineering **Jun 2019 – May 2022**
Gujarat Technological University | ***GPA: 3.8/4.0***
- Relevant Coursework: Machine Learning & AI, Python Programming, Internet of Things, Embedded Systems

KEY TECHNICAL PROJECTS

Real-Time Geospatial Risk Assessment Platform (GitHub)
Python, Flask, PostGIS, JavaScript, Leaflet, OpenWeatherMap, TomTom APIs
- Engineered full-stack web application for urban transportation network risk analysis, integrating live traffic, weather, and spatial data on interactive Leaflet maps for real-time infrastructure threat assessment.
- Built scalable Flask APIs with PostgreSQL/PostGIS enabling city planners to filter, prioritize, and respond to network threats through dynamic multi-source risk scoring algorithms.

Enterprise Document Intelligence System @ VITG Corp.
AWS (Textract, S3, Redshift), Python, Claude LLM, Streamlit
- Automated geospatial data extraction from 2,500+ scanned PDFs/images using AWS Textract and advanced prompt engineering, increasing processing efficiency by 40% for environmental compliance teams.
- Developed production-grade Streamlit dashboards enabling stakeholders to query, visualize, and prioritize conservation projects through AI-powered data insights and interactive mapping.

Autonomous Robot Fleet Monitoring System @ CARDS
Python, ROS, LiDAR, Computer Vision, Dash
- Built comprehensive real-time monitoring dashboard tracking battery health, connectivity, and GPS-based movement for autonomous robot patrols in military simulation environments.
- Integrated OpenCV-based object detection with LIDAR obstacle avoidance, enhancing robotic safety and operational reliability in complex autonomous deployments.

Cross-Modal AI Recommendation Engine (GitHub)
Python, OpenAI API, Computer Vision, Big Data Analytics
- Developed sophisticated recommendation system combining structured data analysis with computer vision-based image understanding, reducing user bounce rate by 20% and increasing engagement by 40%.
- Implemented advanced vision models for automated image style and content analysis, enabling personalized recommendations that significantly improved user experience metrics.

ADDITIONAL TECHNICAL PROJECTS

Cyclists' Performance Analytics Dashboard (GitHub)
Python, Dash, Machine Learning, Statistical Analysis
- Built interactive dashboard for real-time biometric and performance data; predictive analytics improved athlete outcomes by 20%.
- Integrated live data ingestion, model inference, and visualization on a single-stack Dash app.

Edge-AI Mosquito Classification System (Video)
Python, Edge Impulse, TensorFlow Lite, Embedded C
- Co-developed embedded ML model for species identification via wing-beat audio; achieved 94% accuracy with <100 ms inference on ARM Cortex-M.
- Optimized model size to 240 kB, enabling solar-powered field deployment in remote environments.

CERTIFICATIONS & PUBLICATIONS

Professional Certifications

- **AWS Machine Learning Associate Certification** – View Credential
- **AWS Machine Learning Foundations Certification** – Core ML concepts & cloud deployment expertise

Peer-Reviewed Publications

- **Ranjan, R.**, **Shroff, H.**, et al. (2024). *FilletCam AI: Precision fillet color profiling using deep learning. Journal of Agriculture and Food Research.* DOI: 10.1016/j.jafr.2024.101461
- **Trivedi, K.**, & **Shroff, H.** (2021). *Mosquito identification using ML on embedded systems. IEEE ITU Kaleidoscope Conference.* DOI: 10.23919/ITUK53220.2021.9662116